

Banana Problem

Problem 1 – The Quick Courier

Question - In the town of Graphville, there is a quick courier named Lexi who needs to deliver packages to several businesses located in various districts. Graphville is well-connected, but Lexi wants to ensure the route taken is not only efficient but also predictable in order, so her dispatch team can easily track her progress.

Lexi decides to always take the lexicographically smallest path that visits each district exactly once and returns to the starting district. As a budding programmer, you are tasked with helping Lexi find this path using a program.

Input Format- The first line contains an integer n ($2 \leq n \leq 10$), the number of districts.

Each of the next n lines contains n integers, where the j -th integer in the i -th line represents the travel cost from district i to district j . If the number is -1 , it means that the path between the two districts does not exist.

Output Format- Output the lexicographically smallest path (as a sequence of district indices) that visits each district exactly once and returns to the starting district. If the path does not exist, return -1

Sample Test case - Sample Input 1: 3 0 10 15 10 0 20 15 20 0 Sample Output 1: Minimum Cost: 45 Optimal Path: 0 1 2 0 Sample Input 2: 4 0 5 9 10 5 0 6 4 9 6 0 8 10 4 8 0 Sample Output 2: Minimum Cost: 26 Optimal Path: 0 1 3 2 0	Hidden Test Cases - Sample Input 3: 4 0 -1 -1 4 -1 0 3 -1 -1 3 0 2 4 -1 2 0 Sample Output 3: Path does not exist Sample Input 4:4 0 1 15 6 2 0 7 3 9 6 0 12 10 4 8 0 Sample Output 4: Minimum Cost: 21 Optimal Path: 0 1 3 2 0
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	Sample Input 5: 5 0 3 4 2 7 3 0 4 6 3 4 4 0 5 8 2 6 5 0 6 7 3 8 6 0 Sample Output 5: Minimum Cost: 19 Optimal Path: 0 2 1 4 3 0
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